

PPM Framework
Hand-Solved Test Cases
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Internal Technical Reference
July 20, 2026

Contents

Conventions

All cases share these conventions unless explicitly overridden:

- $\gamma = 0.95$ throughout, imposing time value of money in every case. This makes earlier certification strictly preferable and uniquely pins down the optimal allocation timing.
- Earliest certification dates $e_{i,j}$ are set for every case. The default rule is $e_{i,j} = s_i + \lfloor (j - 1) \cdot D_i^{\text{plan}} / M_i \rfloor$, spreading milestones evenly across the planned duration. Any departure is stated explicitly.
- $B_1 = 3 \cdot \sum_i \text{BAC}_i$ (large; cash never binding unless stated).
- $\bar{\eta}_{i,t} = \bar{\eta}$ constant and deterministic unless stated.
- Equal milestone weights $\phi_{i,j} = 1/M_i$ unless stated.
- $\alpha_i = 0, \rho_i = 0$ unless stated.
- $\delta^{\text{rec}} = 0.25, \psi = 0.10$ (FIDIC defaults) unless stated.
- $\pi_i = 0$ unless stated ($CP_i = \text{BAC}_i$).
- S-curve shape $a_i = b_i = 2$ (symmetric Beta) unless stated.
- Monetary units are abstract.

Note: With $\gamma = 0.95$ and earliest certification dates enforced, the optimal strategy in every completing case is to accumulate progress as fast as possible so that $P_{i,t}^{\text{actual}} \geq \theta_{i,j}$ is satisfied by $t = e_{i,j}$, then certify immediately when the window opens. Spending earlier than needed to reach the threshold is wasteful only if it depletes cash needed elsewhere; spending later than $e_{i,j}$ loses discount value. This makes the MILP allocation uniquely determined in all cases below.

1 Group 1: Progress and Milestone Certification

What is being proven: budget converts to progress at rate η , milestones certify at the correct threshold and not before $e_{i,j}$, payments trigger correctly, and milestone ordering is enforced even when progress jumps past multiple thresholds in one period.

1.1 SP-1: Perfect Efficiency, Two Milestones

Economic claim. With $\eta = 1$ and $\gamma = 0.95$, the MILP front-loads spend so that MS1 certifies exactly at $e_{1,1} = 1$ and MS2 certifies exactly at $e_{1,2} = 2$. Spending all budget at $t = 1$ would not accelerate MS2 because $e_{1,2} = 2$ blocks early certification.

Parameters.

$$n = 1, \quad s_1 = 1, \quad f_1 = 4, \quad D^{\text{plan}} = 4, \quad \text{BAC}_1 = 100, \quad CP_1 = 120, \quad \pi_1 = 0.20.$$

$$M_1 = 2, \quad \theta = (0.5, 1.0), \quad \phi = (0.5, 0.5), \quad \bar{\eta} = 1.0.$$

$$e_{1,1} = 1 + \lfloor 0 \cdot 4/2 \rfloor = 1, \quad e_{1,2} = 1 + \lfloor 1 \cdot 4/2 \rfloor = 3.$$

$$\alpha_1 = 0, \quad \rho_1 = 0, \quad \Omega_1 = 1, \quad \mu_1 = 0.30, \quad \tau_1^{\text{tol}} = 2, \quad B_1 = 300, \quad H = 4.$$

Milestone payments.

$$R_{1,1}^{\text{net}} = 60, \quad R_{1,2}^{\text{net}} = 60.$$

Optimal strategy. Spend 50 at $t = 1$ ($P = 0.5 \geq \theta_1 = 0.5$, window $e_{1,1} = 1$ open): MS1 certifies. Spend 50 at $t = 2$ ($P = 1.0 \geq \theta_2 = 1.0$, but $e_{1,2} = 3$ not yet open): progress complete, MS2 pending. At $t = 3$: no spend needed, $e_{1,2} = 3$ opens, MS2 certifies.

t	$x_{1,t}$	$P_{1,t}$	Window open?	Inflow	Net
1	50	0.50	MS1: yes	60	+10
2	50	1.00	MS2: no	0	-50
3	0	1.00	MS2: yes	60	+60
4	0	1.00	—	0	0

Objective.

$$Z^* = \gamma^0(60-50) + \gamma^1(0-50) + \gamma^2(60-0) = 10 - 0.95 \times 50 + 0.9025 \times 60 = 10 - 47.5 + 54.15 = \mathbf{16.65}.$$

Note: Could the MILP defer MS1 spend to $t = 2$ and MS2 spend to $t = 3$? That gives $Z = \gamma^1(60-50) + \gamma^2(60-50) = 0.95 \times 10 + 0.9025 \times 10 = 9.5 + 9.025 = 18.525$. Wait — that is higher. The MILP should defer both. Recalculate: deferring MS1 to $t = 2$ means $P \geq 0.5$ at $t = 2$ with $e_{1,1} = 1$ open. And MS2: spend at $t = 3$ to get $P = 1$ with $e_{1,2} = 3$ open. $Z = \gamma^1(60-50) + \gamma^2(60-50) = 0.95 \times 10 + 0.9025 \times 10 = 18.525$. But then the outflows are also discounted. Actually with $\gamma < 1$ we want inflows early and outflows late — the MILP will actually defer spending to get earlier net.

Let me redo this properly. The MILP maximises:

$$\sum_t \gamma^{t-1} (V_t - x_t).$$

Outflows at t cost γ^{t-1} weight; inflows at t earn γ^{t-1} weight. Since outflows precede inflows (you spend before the milestone certifies), and $\gamma < 1$ discounts future inflows, the net effect is: spend as late as possible while still reaching the threshold by $e_{i,j}$.

For MS1 ($e_{1,1} = 1$, threshold 0.5, requires spend 50 with $\eta = 1$): must spend 50 by $t = 1$. No choice — earliest possible is $t = 1$.

For MS2 ($e_{1,2} = 3$, threshold 1.0, additional spend 50): must have cumulative spend 100 by $t = 3$. Already have 50 from $t = 1$. Need 50 more by $t = 3$. To minimise discounted outflow, spend as late as possible: spend 50 at $t = 3$ itself (just before window opens — but spend and certification happen same period, so spend at $t = 3$, P updates, $e_{1,2} = 3$ satisfied, MS2 certifies at $t = 3$).

Revised optimal strategy.

t	$x_{1,t}$	$P_{1,t}$	Window open?	Inflow	Net
1	50	0.50	MS1: yes	60	+10
2	0	0.50	MS2: no	0	0
3	50	1.00	MS2: yes	60	+10
4	0	1.00	—	0	0

$$Z^* = \gamma^0(60-50) + \gamma^1(0) + \gamma^2(60-50) = 10 + 0 + 0.9025 \times 10 = 10 + 9.025 = \mathbf{19.025}.$$

This is strictly better than the $t = 1, 2$ strategy (16.65). The earliest certification constraint makes spending at $t = 3$ uniquely optimal for MS2 — there is no incentive to spend at $t = 2$ since the window does not open until $t = 3$ and deferring the outflow saves discount.

Assertions for pytest.

- A1. $x_{1,1} = 50 \pm 0.5$, $x_{1,2} = 0 \pm 0.5$, $x_{1,3} = 50 \pm 0.5$, $x_{1,4} = 0$.
- A2. MS1 certified at $t = 1$; MS2 certified at $t = 3$ (not before, since $e_{1,2} = 3$).
- A3. $|Z^* - 19.025| \leq 1.0$.
- A4. Status: `completed`. No τ decrement.

1.2 SP-3: Three Milestones, $\gamma = 0.95$

Economic claim. With $\gamma < 1$ and earliest certification dates, the MILP certifies each milestone exactly when its window opens, spending just enough just in time to meet the progress threshold at each $e_{i,j}$.

Parameters.

$$\begin{aligned} n = 1, \quad s_1 = 1, \quad f_1 = 6, \quad D^{\text{plan}} = 6, \quad \text{BAC}_1 = 60, \quad \pi_1 = 0.50 \Rightarrow CP_1 = 90. \\ M_1 = 3, \quad \theta = (1/3, 2/3, 1), \quad \phi = (1/3, 1/3, 1/3), \quad \bar{\eta} = 1.0. \\ e_{1,1} = 1, \quad e_{1,2} = 1 + \lfloor 1 \cdot 6/3 \rfloor = 3, \quad e_{1,3} = 1 + \lfloor 2 \cdot 6/3 \rfloor = 5. \\ \alpha_1 = 0, \quad \rho_1 = 0, \quad \Omega_1 = 2, \quad \mu_1 = 0.50, \quad \tau_1^{\text{tol}} = 2. \end{aligned}$$

Milestone payments. $R_{1,j}^{\text{net}} = 30$ each.

Optimal strategy. Each milestone requires spend of 20 ($= 60/3$). To minimise discounted outflow: spend 20 at $t = e_{1,j}$ for each j .

t	x	P	Window	Inflow	Net
1	20	1/3	MS1: yes	30	+10
2	0	1/3	—	0	0
3	20	2/3	MS2: yes	30	+10
4	0	2/3	—	0	0
5	20	1	MS3: yes	30	+10
6	0	1	—	0	0

$$Z^* = \gamma^0 \times 10 + \gamma^2 \times 10 + \gamma^4 \times 10 = 10 + 0.9025 \times 10 + 0.8145 \times 10 = 10 + 9.025 + 8.145 = \mathbf{27.17}.$$

Assertions for pytest.

- A1. $x_{1,1} = 20 \pm 0.5$, $x_{1,3} = 20 \pm 0.5$, $x_{1,5} = 20 \pm 0.5$; zero otherwise.
- A2. MS j certified at $t \in \{1, 3, 5\}$ respectively.
- A3. $|Z^* - 27.17| \leq 1.0$.
- A4. Status: `completed`.

1.3 SP-J: Milestone Ordering Enforced on Progress Jump

Economic claim. When a single allocation causes progress to clear both thresholds, both milestones certify in the same period provided both windows are open. If $e_{1,2} > e_{1,1}$, MS2 cannot certify at $t = e_{1,1}$ even though progress qualifies.

Parameters.

$$n = 1, \quad s_1 = 1, \quad f_1 = 4, \quad \text{BAC}_1 = 100, \quad CP_1 = 100, \quad \pi_1 = 0.$$

$$M_1 = 2, \quad \theta = (0.5, 1.0), \quad \phi = (0.5, 0.5), \quad \bar{\eta} = 1.0.$$

$$e_{1,1} = 1, \quad e_{1,2} = 3.$$

$$\alpha_1 = 0, \quad \rho_1 = 0, \quad \Omega_1 = 2, \quad \mu_1 = 0.50, \quad \tau_1^{\text{tol}} = 2, \quad B_1 = 300.$$

Allocation. Spend $x_{1,1} = 100$ at $t = 1$. $P_{1,1} = 1.0$, clearing both $\theta_1 = 0.5$ and $\theta_2 = 1.0$.

MS1: $\theta_1 \leq P$ and $t = 1 \geq e_{1,1} = 1$. Certifies. $R = 50$. MS2: $\theta_2 \leq P$ but $t = 1 < e_{1,2} = 3$. Does *not* certify.

At $t = 3$: no further spend needed. $e_{1,2} = 3$ opens. MS2 certifies. $R = 50$.

Objective.

$$Z^* = \gamma^0(50 - 100) + \gamma^1(0) + \gamma^2(50) = -50 + 0 + 0.9025 \times 50 = -50 + 45.125 = -\mathbf{4.875}.$$

Is there a better strategy? Defer the 100 spend to $t = 3$ (just before both windows are open would be $t = 3$ for MS2): At $t = 3$: spend 100, $P = 1$, both $e_{1,1} = 1$ and $e_{1,2} = 3$ open, both certify. $Z = \gamma^2(100 - 100) = 0$. Better.

Or: spend 50 at $t = 1$ (MS1 at $t = 1$), spend 50 at $t = 3$ (MS2 at $t = 3$): $Z = \gamma^0(50 - 50) + \gamma^2(50 - 50) = 0$. Same.

Or: spend 50 at $t = 1$, spend 50 at $t = 2$ (progress reaches 1.0 at $t = 2$ but $e_{1,2} = 3$ still closed; MS2 certifies at $t = 3$ with no spend): $Z = \gamma^0(50 - 50) + \gamma^1(-50) + \gamma^2(50) = -47.5 + 45.125 = -2.375$. Worse than 0.

MILP optimum: spend nothing until $t = 1$ for MS1 (mandatory since $e_{1,1} = 1$ and we must spend to certify), then spend 50 at $t = 3$ for MS2. $Z^* = 0$.

Revised optimal strategy.

t	x	P	Inflow	Net
1	50	0.50	50	0
2	0	0.50	0	0
3	50	1.00	50	0
4	0	1.00	0	0

$$Z^* = 0 + 0 + 0 + 0 = \mathbf{0} \text{ (all nets are zero so discounting doesn't change it).}$$

Key assertion for ordering: MS2 does not certify at $t = 1$ even though $P_{1,1} = 1.0 \geq \theta_2$ after spending 100 at $t = 1$, because $e_{1,2} = 3 > 1$.

Assertions for pytest.

A1. MS2 not certified at $t = 1$ or $t = 2$ (window closed).

A2. MS2 certified at $t = 3$.

A3. $|Z^* - 0| \leq 1.0$.

A4. Status: `completed` by $t = 3$.

2 Group 2: Discounting and Timing Incentives

What is being proven: $\gamma < 1$ and earliest certification dates together make the optimal allocation uniquely determined: spend just enough just before each $e_{i,j}$ to qualify, and certify the moment the window opens. When two projects compete, complete the one whose milestone payment structure yields higher discounted value first.

2.1 SP-4: Deferring Spend to Last Possible Period is Optimal

Economic claim. With $\gamma < 1$, deferring outflows to the latest period that still allows certification at $e_{i,j}$ is strictly better than front-loading. The margin over the front-loaded strategy is computable.

Parameters.

$$n = 1, \quad s_1 = 1, \quad f_1 = 8, \quad D^{\text{plan}} = 8, \quad \text{BAC}_1 = 100, \quad \pi_1 = 0.40 \Rightarrow CP_1 = 140.$$

$$M_1 = 2, \quad \theta = (0.5, 1), \quad \phi = (0.5, 0.5), \quad \bar{\eta} = 1.0.$$

$$e_{1,1} = 1 + \lfloor 0 \cdot 8/2 \rfloor = 1, \quad e_{1,2} = 1 + \lfloor 1 \cdot 8/2 \rfloor = 5.$$

$$R_{1,j}^{\text{net}} = 70 \text{ each.}$$

Optimal strategy. MS1 requires 50 spend; must be at $t = 1$ (window opens at $t = 1$, must spend to qualify, no benefit deferring since window is already open at $t = 1$). MS2 requires 50 more; window opens at $t = 5$, so spend 50 at $t = 5$.

$$Z^{\text{opt}} = \gamma^0(70 - 50) + \gamma^4(70 - 50) = 20 + 0.95^4 \times 20 = 20 + 0.8145 \times 20 = 20 + 16.29 = \mathbf{36.29}.$$

Front-loaded strategy (spend at $t = 1$ and $t = 2$). MS2 window still closed at $t = 2$; progress qualifies at $t = 2$ but certifies only at $t = 5$.

$$Z^{\text{front}} = \gamma^0(70 - 50) + \gamma^1(-50) + \gamma^4(70) = 20 - 47.5 + 0.8145 \times 70 = 20 - 47.5 + 57.015 = \mathbf{29.515}.$$

$Z^{\text{opt}} = 36.29 > Z^{\text{front}} = 29.515$. Deferring the MS2 spend to $t = 5$ saves $\gamma^1 \times 50 - \gamma^4 \times 50 = 47.5 - 40.725 = 6.775$ in discounted outflow with identical inflow.

Assertions for pytest.

A1. $x_{1,1} = 50 \pm 0.5$.

A2. $x_{1,2} = x_{1,3} = x_{1,4} = 0 \pm 0.5$.

A3. $x_{1,5} = 50 \pm 0.5$.

A4. MS1 certified at $t = 1$; MS2 certified at $t = 5$.

A5. $Z^* \geq 35.5$.

A6. $Z^* > 29.5$ (front-loaded strategy strictly dominated).

2.2 MP-K: Two Projects, Front-Heavy vs. Back-Heavy Milestones

Economic claim. Two projects identical in BAC, CP , η , and duration differ only in milestone weight distribution. Under $\gamma < 1$, completing the front-heavy project first (large early payment at $e_{A,1}$) yields higher discounted inflow than completing the back-heavy project first.

Parameters.

$$n = 2, \quad s_i = 1, \quad f_i = 6, \quad D^{\text{plan}} = 6, \quad \text{BAC}_i = 60, \quad \pi_i = 0.50 \Rightarrow CP_i = 90.$$

$$M_i = 2, \quad \theta_i = (0.5, 1.0), \quad \bar{\eta}_i = 1.0.$$

$$e_{i,1} = 1, \quad e_{i,2} = 1 + \lfloor 1 \cdot 6/2 \rfloor = 4.$$

$$\phi_A = (0.70, 0.30), \quad \phi_B = (0.30, 0.70).$$

$$R_{A,1}^{\text{net}} = 63, \quad R_{A,2}^{\text{net}} = 27, \quad R_{B,1}^{\text{net}} = 27, \quad R_{B,2}^{\text{net}} = 63.$$

Each milestone requires spend of 30. $B_1 = 300$.

Strategy 1 (A then B): complete A first. $t = 1$: spend 30 on A (MS1_A at $e_{A,1} = 1$), 30 on B (MS1_B at $e_{B,1} = 1$). Both MS1 certify simultaneously. $t = 4$: spend 30 on A (MS2_A at $e_{A,2} = 4$), 30 on B (MS2_B at $e_{B,2} = 4$). Both MS2 certify simultaneously.

Both projects complete at the same time in this symmetric allocation. The weights affect inflows but not timing here.

$$Z_{\text{sim}} = \gamma^0(63 + 27 - 30 - 30) + \gamma^3(27 + 63 - 30 - 30) = 30 + 0.857 \times 30 = 30 + 25.72 = \mathbf{55.72}.$$

Can we do better by sequencing? With $B_1 = 300$ (non-binding), can spend 30+30 at $t = 1$ and 30+30 at $t = 4$ regardless. Sequencing does not help since both MS1 windows open at $t = 1$ and both MS2 windows at $t = 4$.

The test here is not about ordering but about weights: A's MS1 payment (63) is larger than B's MS1 payment (27). Under $\gamma < 1$, if we could stagger, getting A's MS1 earlier would help. But since both $e_{i,1} = 1$, the simultaneity is forced.

Adjust: set $e_{A,1} = 1$, $e_{B,1} = 2$ to force staggering:

t	x_A	x_B	Inflow	Net
1	30	0	63 (MS1 _A)	+33
2	0	30	27 (MS1 _B)	-3
4	30	30	27+63=90	+30

$$Z_{A1\text{st}} = \gamma^0(63 - 30) + \gamma^1(27 - 30) + \gamma^3(90 - 60) = 33 - 2.85 + 0.857 \times 30 = 33 - 2.85 + 25.72 = \mathbf{55.87}.$$

$$\text{Reverse (B's MS1 first at } t = 1, \text{ A's at } t = 2): Z_{B1\text{st}} = \gamma^0(27 - 30) + \gamma^1(63 - 30) + \gamma^3(90 - 60) = -3 + 0.95 \times 33 + 25.72 = -3 + 31.35 + 25.72 = \mathbf{54.07}.$$

$$Z_{A1\text{st}} = 55.87 > Z_{B1\text{st}} = 54.07. \text{ A first dominates.}$$

Use $e_{A,1} = 1$, $e_{B,1} = 2$ in this case to isolate the ordering decision.

Assertions for pytest.

A1. MS1_A certified at $t = 1$; MS1_B certified at $t = 2$.

A2. $Z^* \geq 55.5$.

A3. $Z^* > 54.07$ (B-first strictly dominated).

A4. Both projects completed.

3 Group 3: Advance Payment and Recovery Mechanics

What is being proven: advance arrives at $t = s_i$, recovery deductions apply once ψ_i is crossed, deductions stop once advance is fully recovered, and the payment identity holds exactly. With $\gamma < 1$, the MILP prefers to receive the advance early and defer outflows.

3.1 SP-2: Advance Harvest, Efficiency Decay, Forced Termination

Economic claim. A large advance combined with rapidly decaying efficiency makes completion economically impossible. The MILP takes the advance and accepts termination. Settlement exactly cancels the advance.

Parameters.

$$n = 1, \quad s_1 = 1, \quad f_1 = 6, \quad \text{BAC}_1 = 100, \quad \text{CP}_1 = 100, \quad \pi_1 = 0.$$

$$M_1 = 2, \quad \theta = (0.5, 1.0), \quad \phi = (0.5, 0.5), \quad e_{1,1} = 1, \quad e_{1,2} = 4.$$

$$\alpha_1 = 0.30 \Rightarrow A_1 = 30, \quad \delta^{\text{rec}} = 0.25, \quad \psi = 0.10, \quad \rho_1 = 0.$$

$$\Omega_1 = 1, \quad \mu_1 = 0.05, \quad \tau_1^{\text{tol}} = 1.$$

$$\bar{\eta}_{1,1} = 1.00, \quad \bar{\eta}_{1,t} = 0.05 \text{ for } t \geq 2.$$

Zero spend: counter $1 \rightarrow 0$ at $t = 1$, termination at $t = 2$. $P = 0$. $R^{\text{term}} = -30$.
 $Z^* = \gamma^0 \cdot 30 + \gamma^1 \cdot (-30) = 30 - 28.5 = \mathbf{1.5}$.

Note: With $\gamma < 1$, the advance inflow at $t = 1$ is worth more than the settlement outflow at $t = 2$ (discounted). This gives $Z^* = 1.5 > 0$, unlike the $\gamma = 1$ case where $Z^* = 0$. The discounting actually makes the advance harvest slightly profitable even when settlement cancels it in nominal terms.

Assertions for pytest.

A1. $\sum_t x_{1,t} \leq 1.0$.

A2. Project terminates at $t \leq 3$.

A3. $R_1^{\text{term}} = -30 \pm 1$.

A4. $|Z^* - 1.5| \leq 0.5$.

3.2 SP-A: Late Recovery Threshold ($\psi = 0.40$)

Parameters and payment mechanics unchanged from previous version.

$$n = 1, \quad \text{BAC}_1 = 120, \quad \pi_1 = 0.50 \Rightarrow \text{CP}_1 = 180, \quad M_1 = 3, \quad \phi = (1/3, 1/3, 1/3).$$

$$\alpha_1 = 0.20 \Rightarrow A_1 = 36, \quad \psi = 0.40, \quad \delta^{\text{rec}} = 0.30, \quad \rho_1 = 0, \quad \bar{\eta} = 1.0.$$

$$e_{1,1} = 1, \quad e_{1,2} = 3, \quad e_{1,3} = 5, \quad f_1 = 6.$$

$$R_{1,1}^{\text{net}} = 60, \quad R_{1,2}^{\text{net}} = 42, \quad R_{1,3}^{\text{net}} = 42.$$

Spend 40 at each $e_{1,j}$:

$$Z^* = \gamma^0(36+60-40) + \gamma^2(42-40) + \gamma^4(42-40) = 56 + 0.9025 \times 2 + 0.8145 \times 2 = 56 + 1.805 + 1.629 = \mathbf{59.43}.$$

Assertions for pytest.

A1. $R_{1,1}^{\text{net}} = 60 \pm 0.5$.

A2. $R_{1,2}^{\text{net}} = 42 \pm 0.5$.

A3. $R_{1,3}^{\text{net}} = 42 \pm 0.5$.

A4. $x_{1,1} = 40 \pm 0.5, x_{1,3} = 40 \pm 0.5, x_{1,5} = 40 \pm 0.5$.

A5. $|Z^* - 59.43| \leq 1.5$.

3.3 SP-B: Advance Fully Recovered Before Final Milestone

Parameters.

$$\text{BAC}_1 = 60, \quad \pi_1 = 0.50 \Rightarrow CP_1 = 90, \quad M_1 = 3, \quad \phi = (1/3, 1/3, 1/3).$$

$$\alpha_1 = 0.20 \Rightarrow A_1 = 18, \quad \psi = 0.10, \quad \delta^{\text{rec}} = 0.30, \quad \rho_1 = 0.$$

$$f_1 = 4, \quad e_{1,1} = 1, \quad e_{1,2} = 2, \quad e_{1,3} = 3.$$

$$R_{1,1}^{\text{net}} = 21, \quad R_{1,2}^{\text{net}} = 21, \quad R_{1,3}^{\text{net}} = 30.$$

Spend 20 at each $e_{1,j}$:

$$Z^* = \gamma^0(18+21-20) + \gamma^1(21-20) + \gamma^2(30-20) = 19 + 0.95 \times 1 + 0.9025 \times 10 = 19 + 0.95 + 9.025 = \mathbf{28.975}.$$

Assertions for pytest.

$$\text{A1. } R_{1,3}^{\text{net}} = 30 \pm 0.5 \text{ (no deduction on MS3).}$$

$$\text{A2. } x_{1,1} = 20 \pm 0.5, \quad x_{1,2} = 20 \pm 0.5, \quad x_{1,3} = 20 \pm 0.5.$$

$$\text{A3. } |Z^* - 28.975| \leq 1.0.$$

4 Group 4: Retention

4.1 SP-C: Retention Only, Lump Release at Completion

Parameters.

$$\text{BAC}_1 = 60, \quad \pi_1 = 0.50 \Rightarrow CP_1 = 90, \quad M_1 = 3, \quad \rho_1 = 0.10.$$

$$f_1 = 6, \quad e_{1,1} = 1, \quad e_{1,2} = 3, \quad e_{1,3} = 5.$$

$$R_{1,j}^{\text{net}} = 27 \text{ each; } R_1^{\text{ret}} = 9 \text{ at } t = e_{1,3} = 5.$$

Final period inflow at $t = 5$: $27 + 9 = 36$.

$$Z^* = \gamma^0(27-20) + \gamma^2(27-20) + \gamma^4(36-20) = 7 + 0.9025 \times 7 + 0.8145 \times 16 = 7 + 6.318 + 13.032 = \mathbf{26.35}.$$

Assertions for pytest.

$$\text{A1. } R_{1,j}^{\text{net}} = 27 \pm 0.5 \text{ for } j = 1, 2, 3.$$

$$\text{A2. } R_1^{\text{ret}} = 9 \pm 0.5, \text{ released only at } t = 5.$$

$$\text{A3. } x_{1,1} = 20 \pm 0.5, \quad x_{1,3} = 20 \pm 0.5, \quad x_{1,5} = 20 \pm 0.5.$$

$$\text{A4. } |Z^* - 26.35| \leq 1.0.$$

4.2 SP-D: Retention and Advance Recovery Simultaneously Active

Parameters.

$$\text{BAC}_1 = 60, \quad \pi_1 = 0.50 \Rightarrow CP_1 = 90, \quad M_1 = 3, \quad \rho_1 = 0.10.$$

$$\alpha_1 = 0.20 \Rightarrow A_1 = 18, \quad \psi = 0.10, \quad \delta^{\text{rec}} = 0.30.$$

$$f_1 = 6, \quad e_{1,1} = 1, \quad e_{1,2} = 3, \quad e_{1,3} = 5.$$

$$R_{1,1}^{\text{net}} = 18, \quad R_{1,2}^{\text{net}} = 18, \quad R_{1,3}^{\text{net}} = 27; \quad R_1^{\text{ret}} = 9.$$

$$Z^* = \gamma^0(18+18-20) + \gamma^2(18-20) + \gamma^4(27+9-20) = 16 + 0.9025 \times (-2) + 0.8145 \times 16 = 16 - 1.805 + 13.032 = \mathbf{27.227}.$$

Assertions for pytest.

- A1. $R_{1,1}^{\text{net}} = 18 \pm 0.5$, $R_{1,2}^{\text{net}} = 18 \pm 0.5$.
- A2. $R_{1,3}^{\text{net}} = 27 \pm 0.5$ (retention only on MS3).
- A3. $R_1^{\text{ret}} = 9 \pm 0.5$ at $t = 5$.
- A4. $|Z^* - 27.227| \leq 1.0$.

5 Group 5: Termination Conditions in Isolation

Note: $\text{CPI}_{i,t} = \eta_{i,t}$ always. With $\eta = 1$, EAC = BAC and Condition 2 cannot fire. With $\eta < 1$ and fast planned S-curve, both conditions fire. These cases verify counter independence; Z^* values are not the focus and are omitted where the project does not complete.

5.1 SP-E: Only Condition 1 Fires

Parameters unchanged. $\eta = 1$, spend below plan rate. Counter stays at $\tau^{\text{tol}} = 3$. $e_{1,1}$ not relevant (MS never certifies in the observation window). Focus is counter behavior only.

Assertions for pytest.

- A1. $\text{SPI}_{1,t} < 1$ for $t \geq 3$.
- A2. $\text{CPI}_{1,t} = 1.0 \pm 0.01$ for all t .
- A3. $\tau_{1,t}^{\text{rem}} = 3$ for all t .
- A4. Status never `terminated`.

5.2 SP-F: Only Condition 2 Fires

Parameters unchanged. $\eta = 0.5$, long plan, large spend. Counter stays at $\tau^{\text{tol}} = 3$.

Assertions for pytest.

- A1. $\text{SPI}_{1,t} > 1$ for $t \leq 10$.
- A2. $\text{EAC}_{1,t} > 105$ from $t = 1$.
- A3. $\tau_{1,t}^{\text{rem}} = 3$ for all observed periods.
- A4. Status never `terminated`.

5.3 SP-G: Both Fire, Condition 1 Lapses, Counter Resets

Parameters unchanged. Efficiency schedule: $\bar{\eta}_t = 0.3$ for $t = 1, 2$; $\bar{\eta}_3 = 1.0$; $x_3 = 80$.
Counter trajectory: $\tau = [3, 2, 3, \dots]$ at $t = 1, 2, 3$.

Assertions for pytest.

- A1. $\tau_{1,1}^{\text{rem}} = 3$.
- A2. $\tau_{1,2}^{\text{rem}} = 2$.
- A3. $\tau_{1,3}^{\text{rem}} = 3$ (reset).
- A4. Status at $t = 3$: `active`.

6 Group 6: Cure Period and Termination Execution

6.1 SP-5: Zero Efficiency, Exact Counter Boundary

Parameters unchanged: $\eta = 0$, $\tau^{\text{tol}} = 3$, $A_1 = 20$. Termination at $t = 4$.

$$Z^* = \gamma^0 \cdot 20 + \gamma^3 \cdot (-20) = 20 - 0.95^3 \times 20 = 20 - 0.8574 \times 20 = 20 - 17.15 = \mathbf{2.85}.$$

With $\gamma < 1$, the advance inflow at $t = 1$ is worth more than the discounted settlement at $t = 4$, giving a small positive Z^* .

Assertions for pytest.

- A1. $\tau = [2, 1, 0]$ at $t = 1, 2, 3$.
- A2. Termination at $t = 4$.
- A3. $R_1^{\text{term}} = -20 \pm 0.5$.
- A4. $|Z^* - 2.85| \leq 0.5$.

6.2 SP-H: Partial Progress at Termination, Positive Settlement

Parameters unchanged: $\eta = 0.5$, $A_1 = 10$, $\tau^{\text{tol}} = 2$, $\theta_1 = 0.30$ with $e_{1,1} = 1$. Spend 60 at $t = 1$, MS1 certifies ($R = 30$, advance recovered). Termination at $t = 3$. $R^{\text{term}} = 30$.

$$Z^* = \gamma^0(10 + 30 - 60) + \gamma^1(0) + \gamma^2(30) = -20 + 0 + 0.9025 \times 30 = -20 + 27.075 = \mathbf{7.075}.$$

Zero-spend comparison: $Z = \gamma^0(10) + \gamma^1(-10) = 10 - 9.5 = 0.5$. MILP prefers spending 60: $7.075 > 0.5$. ✓

Assertions for pytest.

- A1. MS1 certified at $t = 1$.
- A2. $R_1^{\text{term}} = 30 \pm 0.5$.
- A3. Termination at $t = 3$.
- A4. $|Z^* - 7.075| \leq 1.0$.

6.3 SP-I: Cure Period Interrupted and Restarted

Parameters unchanged. Counter trajectory: $[3, 2, 3, 2, 1, 0]$ at $t = 1 \dots 6$, termination at $t = 7$.

Z^* not computed (no milestones certify, no advance). Focus is counter trajectory assertion only.

Assertions for pytest.

- A1. $\tau_{1,2}^{\text{rem}} = 2$.
- A2. $\tau_{1,3}^{\text{rem}} = 3$ (reset).
- A3. $\tau_{1,6}^{\text{rem}} = 0$.
- A4. Termination at $t = 7$.

7 Group 7: Portfolio Cash Balance and Coupling

7.1 MP-3: Advance Harvest Funds Concurrent Good Project

Same as before with $B_1 = 90$, $\gamma = 0.95$.

$$Z^* = \gamma^0(30+60-40) + \gamma^1(60-40) + \gamma^2(-30) = 50 + 0.95 \times 20 + 0.9025 \times (-30) = 50 + 19 - 27.075 = \mathbf{41.925}.$$

Assertions for pytest.

- A1. $B_t \geq 0$ for all t .
- A2. Good completed by $t = 2$.
- A3. $|Z^* - 41.925| \leq 1.5$.

7.2 MP-A: Sequential Funding

Same parameters, $B_1 = 30$. With $\gamma = 0.95$:

Total inflow = 180, total outflow = 120 (unchanged). Z^* depends on timing. Inflows at $t = 1, 2, 3, 4$; outflows at $t = 1, 2, 3, 4$. Exact timing requires tracing the balance constraints; the key assertion is $x_{2,1} = 0$ and $Z^* < 60$ (discounting reduces it relative to the $\gamma = 1$ case).

Assertions for pytest.

- A1. $x_{2,1} = 0$.
- A2. $B_t \geq 0$ for all t .
- A3. Both completed.
- A4. $Z^* \geq 50$ (discounted value; exact depends on timing).

7.3 MP-B: Negative Settlement Drains Shared Balance

Same parameters. Z^* with $\gamma = 0.95$:

$$Z^* = \gamma^0(18-30+45) + \gamma^1(45-30) + \gamma^2(-18) = 33 + 0.95 \times 15 + 0.9025 \times (-18) = 33 + 14.25 - 16.245 = \mathbf{31.005}.$$

Assertions for pytest.

- A1. $R_{\text{Term}}^{\text{term}} = -18 \pm 0.5$.
- A2. $B_4 = 130 \pm 1$.
- A3. $|Z^* - 31.005| \leq 1.5$.

8 Group 8: Portfolio Selection Under Constraints

8.1 MP-2: Good vs. Bad, Unconstrained Cash

Good project: MS1 at $e = 1$, MS2 at $e = 5$ (using default rule, $f = 8$, $M = 2$: $e_2 = 1 + \lfloor 4 \rfloor = 5$).
Front-load: spend 40 at $t = 1$ (MS1), 40 at $t = 5$ (MS2).

$$Z_G = \gamma^0(60 - 40) + \gamma^4(60 - 40) = 20 + 0.8145 \times 20 = 20 + 16.29 = \mathbf{36.29}.$$

$$\text{Bad: } Z_B = \gamma^0 \cdot 30 + \gamma^1(-30) = 30 - 28.5 = 1.5 \text{ (same as SP-2). } Z^* = 36.29 + 1.5 = \mathbf{37.79}.$$

Assertions for pytest.

- A1. $\sum_t x_{\text{Bad},t} \leq 1.0$.
- A2. Bad terminates at $t \leq 4$.
- A3. Good completed.
- A4. $|Z^* - 37.79| \leq 1.5$.

8.2 MP-4: Three Projects, Low Always Terminates

Z^* with $\gamma = 0.95$, milestones certifying at $t = 1$ and $t = 2$ (same timing as before):

$$Z^* = \gamma^0(45 + 36 - 30 - 30) + \gamma^1(45 + 36 - 30 - 30) + 0 = 21 + 0.95 \times 21 = 21 + 19.95 = \mathbf{40.95}.$$

Assertions for pytest.

- A1. High and Mid completed by $t = 2$.
- A2. Low terminated.
- A3. $|Z^* - 40.95| \leq 1.5$.
- A4. $B_t \geq 0$.

8.3 MP-C: Cash Sufficient for Only One Project

$B_1 = 60$, Hi completes with Z_{Hi} depending on e dates. Default $e_{\text{Hi},1} = 1$, $e_{\text{Hi},2} = 1 + \lfloor 2 \rfloor = 3$ ($f = 4$, $M = 2$, $D = 4$: $e_2 = 1 + 2 = 3$).

Spend 30 at $t = 1$ (MS1, $R = 45$), 30 at $t = 3$ (MS2, $R = 45$):

$$Z_{\text{Hi}} = \gamma^0(45 - 30) + \gamma^2(45 - 30) = 15 + 0.9025 \times 15 = 15 + 13.54 = \mathbf{28.54}.$$

$$Z^* = 28.54.$$

Assertions for pytest.

- A1. Lo terminated.
- A2. Hi completed.
- A3. $|Z^* - 28.54| \leq 1.5$.
- A4. $B_t \geq 0$.

8.4 MP-D: Bridge Cash Unlocks Major Projects

Same structure. With $\gamma = 0.95$ the option value of Bridge is slightly reduced (future inflows from Alpha/Beta discounted) but still positive. Key assertion remains $Z^* > Z_{\text{Alphaalone}}$.

Assertions for pytest.

- A1. Bridge completed.
- A2. $Z^* > 28$ (better than completing only one major project at discounted value).
- A3. $B_t \geq 0$.

9 Group 9: Horizon and Staggered Starts

9.1 MP-1: Two Non-Overlapping Projects

A: $e_{A,1} = 1$, $e_{A,2} = 3$. B: $e_{B,1} = 5$, $e_{B,2} = 8$. $Z_A = \gamma^0 \times 10 + \gamma^2 \times 10 = 10 + 9.025 = 19.025$. $Z_B = \gamma^4 \times 10 + \gamma^7 \times 10 = 8.145 + 6.983 = 15.128$. $Z^* = 19.025 + 15.128 = \mathbf{34.153}$.

Assertions for pytest.

A1. $H = 10$.

A2. $x_{A,t} = 0$ for $t \geq 5$; $x_{B,t} = 0$ for $t \leq 4$.

A3. $|Z^* - 34.153| \leq 1.5$.

9.2 MP-E: Three Projects with Staggered Starts

P1: $s = 1$, $f = 4$, $e_{1,1} = 1$, $e_{1,2} = 3$. P2: $s = 4$, $f = 8$, $e_{2,1} = 4$, $e_{2,2} = 6$. P3: $s = 9$, $f = 14$, $e_{3,1} = 9$, $e_{3,2} = 11$.

Each: $Z_i = \gamma^{e_{i,1}-1} \times 15 + \gamma^{e_{i,2}-1} \times 15$.

$Z_1 = \gamma^0 \times 15 + \gamma^2 \times 15 = 15 + 13.54 = 28.54$. $Z_2 = \gamma^3 \times 15 + \gamma^5 \times 15 = 12.86 + 10.97 = 23.83$. $Z_3 = \gamma^8 \times 15 + \gamma^{10} \times 15 = 9.30 + 7.94 = 17.24$. $Z^* = 28.54 + 23.83 + 17.24 = \mathbf{69.61}$.

Assertions for pytest.

A1. $H = 14$.

A2. $x_{1,t} = 0$ for $t \geq 5$; $x_{2,t} = 0$ for $t \leq 3$ and $t \geq 9$; $x_{3,t} = 0$ for $t \leq 8$.

A3. Spend at $e_{i,j}$ for each project.

A4. $|Z^* - 69.61| \leq 2.0$.

10 Summary Table

Case	Group	Pattern tested	Outcome	Z^*	Key assertion
SP-1	G1	$\eta = 1$, e pins timing	Completed	19.03	Spend at $t = 1, 3$; MS2 blocked until $e = 3$
SP-3	G1	3 MS, spend at each $e_{i,j}$	Completed	27.17	Spend at $t = 1, 3, 5$
SP-J	G1	Progress jump; e blocks MS2	Completed	0	MS2 not certified until $t = 3$
SP-4	G2	Defer spend to $e_{i,2} = 5$	Completed	36.29	$Z^* > 29.5$ (front-load dominated)
MP-K	G2	A-first beats B-first	Both comp.	55.87	$Z > 54.07$; A's large MS1 first
SP-2	G3	Advance harvest; discount gain	Terminated	1.5	$Z > 0$ due to $\gamma < 1$
SP-A	G3	Late ψ ; MS1 full	Completed	59.43	MS1 full; MS2,3 deducted
SP-B	G3	Recovery done before MS3	Completed	28.975	MS3 full; $R = 30$

Case	Group	Pattern tested	Outcome	Z^*	Key assertion
SP-C	G4	Retention; lump re- lease	Completed	26.35	Release at $t = 5$; $R^{\text{ret}} = 9$
SP-D	G4	Retention + advance	Completed	27.23	MS3 retention only
SP-E	G5	Only Cond. 1; counter fixed	Active	—	$\tau = 3$ always
SP-F	G5	Only Cond. 2; counter fixed	Active	—	$\tau = 3$ always
SP-G	G5	Reset mid-cure	Active	—	$\tau = [3, 2, 3]$
SP-5	G6	$\eta = 0$; counter boundary	Terminated	2.85	$\tau = [2, 1, 0]$; $Z > 0$ due to γ
SP-H	G6	Partial progress; $R^{\text{term}} > 0$	Terminated	7.075	$R^{\text{term}} = 30$; spend pre- ferred
SP-I	G6	Cure reset; late ter- mination	Terminated	—	Term at $t = 7$ not $t = 4$
MP-3	G7	Advance funds con- current project	Mixed	41.93	$B_t \geq 0$; discounted Z
MP-A	G7	Sequential funding	Both comp.	≥ 50	$x_{2,1} = 0$
MP-B	G7	Negative settlement drains B	Mixed	31.01	$B_4 = 130$
MP-2	G8	Good vs. Bad	Bad term.	37.79	Good completes; Bad harvested
MP-4	G8	Low terminates	Low term.	40.95	H&M complete; Low terminates
MP-C	G8	One project's worth of cash	Lo term.	28.54	Hi completes; Lo starved
MP-D	G8	Bridge unlocks ma- jors	All comp.	> 28	Option value realised
MP-1	G9	Non-overlapping; additive Z	Both comp.	34.15	$Z = Z_A + Z_B$ dis- counted
MP-E	G9	Staggered starts; $H = 14$	All comp.	69.61	Spend at each $e_{i,j}$; windows respected

Table 1: All test cases with $\gamma = 0.95$ and earliest certification dates. Z^* values reflect discounting; cases marked — have no meaningful Z^* (focus is counter behavior).